Concentration for Linear Regression

$$y_i = a^T x_i + n_i$$

$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ are given

x_k could be chosen as a function of $(x_1, y_1), \dots, (x_{k-1}, y_{k-1})$. Let $\mathcal{F}_{k-1} = \big((x_1, y_1), \dots, (x_{k-1}, y_{k-1}), x_k\big)$

$n_1, \dots, n_{k-1}$

$$\mathbb{E}\left(e^{\theta n_k} \middle| \mathcal{F}_{k-1}\right) \le e^{\frac{\theta^2}{2}} \quad \text{(subG-1 property)}$$

$\hat{a}$ is chosen to solve

$$\min \frac{1}{2} \sum_{i=1}^{n} \left(y_i - a^T x_i\right)^2 + \frac{\lambda \|a\|^2}{2}$$

$$-\sum_{i=1}^{n} (y_i - a^T x_i) x_i + \lambda a = 0$$

$$-\sum_{i=1}^{n} x_i y_i + \left(\sum_{i=1}^{n} x_i x_i^T + \lambda I\right) a = 0$$

$$\hat{a}_n = \underbrace{\left(\lambda I + \underbrace{\sum_{i=1}^{n} x_i x_i^T}_{V_n}\right)^{-1}}_{V_n} \sum_{i=1}^{n} x_i y_i$$

$$\hat{a}_n - a = V_n^{-1} \sum_{i=1}^{n} x_i (a^T x_i + n_i) - a$$

$$= \left(V_n^{-1} \sum_{i=1}^{n} x_i x_i^T - I\right) a + V_n^{-1} \sum_{i=1}^{n} x_i n_i$$

$$= \left(V_n^{-1} (V_n - \lambda I) - I\right) a + V_n^{-1} \sum_{i=1}^{n} x_i n_i$$

$$= V_n^{-1} \sum_{i=1}^{n} x_i n_i$$

$$= \left(V_n \left(y_n - \right.\right.$$

$$= -\lambda V_n^{-1} a + V_n^{-1} \sum_{i=1}^{n} x_i n_i$$

$$\|a_n - \hat{a}\|_{V_n} \le \lambda \|V_n^{-1} a\|_{V_n} + \left\| V_n^{-1} \sum_{i=1}^{n} x_i n_i \right\|_{V_n}$$

$$= \lambda \sqrt{a^T V_n^{-1} a} + \left\| \sum_{i=1}^{n} x_i n_i \right\|_{V_n^{-1}}$$

since $V_n \ge \lambda I$

$$\le \sqrt{\lambda}\, \|a\| + \underbrace{\left\| \sum_{i=1}^{n} x_i n_i \right\|_{V_n^{-1}}}_{S_n}$$

Let $M_n(\theta) = e^{\theta^T S_n - \frac{1}{2}\|\theta\|^2_{\tilde{V}_n}}$

$$E\left(M_k(\theta) \mid F_{k-1}\right)$$ where $F_{k-1} = (x_1, y_1), \ldots, (x_{k-1}, y_{k-1}), n_1, \ldots, n_{k-1}, x_k$

$$= E\left(e^{\theta^T (S_{k-1} + x_k n_k) - \frac{1}{2} \theta^T (\tilde{V}_{k-1} + x_k x_k^T)\theta} \,\middle|\, F_{k-1} \right)$$

$$= M_{k-1}(\theta)\, E\left(e^{\theta^T x_k n_k - \frac{1}{2}\|\theta\|^2 x_k x_k^T} \,\middle|\, F_{k-1} \right)$$

$$\le M_{k-1}(\theta)\, e^{\frac{1}{2}\|\theta^T x_k\|^2 - \frac{1}{2}\|\theta\|^2 x_k x_k^T}$$

↑ subG property of n_k

$$= M_{k-1}(\theta) \Rightarrow \text{supermartingale!}$$

Let $\bar{M}_t = E\left(M_t(\theta)\right)$, $\theta \sim N(0, \lambda^{-1} I)$

$$E\left(\bar{M}_t \mid F_{t-1}\right) = E\left(\int M_t(\theta) f(\theta) d\theta \mid F_{t-1} \right)$$

$$\mathbb{E}\left(\bar{M}_t \mid F_{t-1}\right) = \mathbb{E}\left(\int_\theta M_t(\theta)\, f(\theta)\, d\theta \mid F_{t-1}\right)$$

$$= \int_\theta \mathbb{E}\left(M_t(\theta) \mid F_{t-1}\right) f(\theta)\, d\theta$$

$$\leq \int_\theta M_{t-1}(\theta)\, f(\theta)\, d\theta$$

$$= \bar{M}_{t-1}(\theta)$$

$$\bar{M}_t = \left(\frac{\lambda^d}{\det(V_t)}\right)^{\frac{1}{2}} \exp\left(\frac{1}{2}\|S_t\|^2_{V_t^{-1}}\right)$$

$$= \exp\left(\frac{1}{2}\|S_t\|^2_{V_t^{-1}} - \frac{1}{2}\log\frac{\det(V_t)}{\lambda^d}\right)$$

we are interested in the prob. that this exceeds $\frac{1}{\delta}$ for $t \in \{1, 2, \ldots, T\}$

$$\Rightarrow P\left(\sup_{t\geq 0} \bar{M}_t \geq \frac{1}{\delta}\right) \leq \frac{1}{\delta} \quad \left(= \frac{\mathbb{E}(\bar{M}_0)}{\frac{1}{\delta}} = \frac{1}{\frac{1}{\delta}}\right)$$

↑ maximal inequality

$\bar{M}_0$ can be taken to be 1 since $\mathbb{E}(\bar{M}_1) \leq 1$

Let $\frac{1}{\delta} = \varepsilon$

w.p. $\geq 1-\varepsilon$,

$$\frac{1}{2}\|S_t\|^2_{V_t^{-1}} \leq \frac{1}{2}\log\frac{\det(V_t)}{\lambda^d} + \log\frac{1}{\varepsilon} \quad \forall t \geq 1$$

Or

$$\|\theta_t\| + \sqrt{\log\frac{\det(V_t)}{\lambda^d} + 2\log\frac{1}{\varepsilon}}$$

Or

$$\|a-\hat{a}_t\|_{V_t} \le \sqrt{\lambda}\|a\| + \sqrt{\log\frac{\det(V_t)}{\lambda^d} + 2\log\frac{1}{\varepsilon}}$$

w.p. $\ge 1-\varepsilon$ $\forall t \ge 1$

$\Leftrightarrow$ If $\lambda = 1$:

$$\|S_t\|^2_{V_t^{-1}} \le \log\frac{\det(V_t)}{\varepsilon^2} \qquad \forall t \ge 1$$

and

$$\|a-\hat{a}\|_{V_t} \le \|a\| + \sqrt{\log\frac{\det(V_t)}{\varepsilon^2}}$$

w.p. $\ge 1-\varepsilon$

Both of these results will be useful later.

Intuition behind defining $M_t(\theta)$ and $\bar{M}_t$

We are interested in

$$P\left(\|a_n-\hat{a}\|_{V_n} \ge \delta\right),$$ but it is difficult to bound this directly since it is difficult to get a handle on

$$E\left(e^{\theta\|a_n-\hat{a}\|_{V_n}}\right)$$

But, notice that when $\lambda = 0$

$$\frac{1}{2}\|\hat{a}_t - a\|^2_{V_t} = \max_u \; u^T S_t - \frac{1}{2}\|u\|^2_{V_t}$$

To see this, note that the

$$\text{LHS} = \frac{1}{2}(\hat{a}-a)^T V_t (\hat{a}-a)$$

Recall $\hat{a}_t - a = V_t^{-1} S_t$ when $\lambda = 0$

$$\Rightarrow \quad \text{LHS} = \frac{1}{2}(V_t^{-1} S_t)^T V_t (V_t^{-1} S_t)$$

$$= \frac{1}{2} S_t^T V_t^{-1} S_t$$

RHS: $\quad S_t - V_t u = 0 \Rightarrow u = V_t^{-1} S_t$

$$\Rightarrow \quad \text{RHS} = S_t^T V_t^{-1} S_t - \frac{1}{2} S_t^T V_t^{-1} S_t$$

$$= \frac{1}{2} S_t^T V_t^{-1} S_t$$

$\Rightarrow$ RHS = LHS when $\lambda = 0$

Let's consider the problem when $\lambda = 0$.

$$P\left(\|\hat{a}_t - a\|^2_{\tilde{V}_t} \geq \varepsilon \right)$$

$$= P\left(\max_u \; u^T S_t - \frac{1}{2}\|u\|^2_{\tilde{V}_t} \geq \varepsilon \right)$$

$$= P\left(e^{\max_u \; u^T S_t - \frac{1}{2}\|u\|^2_{\tilde{V}_t}} \geq e^{\varepsilon} \right)$$

$$= P\left(\max_u \; e^{u^T S_t - \frac{1}{2}\|u\|^2_{\tilde{V}_t}} \geq e^{\varepsilon} \right) \quad (*)$$

- We can try to apply the maximal inequality but we want to get rid of the assumption

- We can try to apply the maximal inequality but we want to get rid of the assumption $\lambda > 0$, so we define $M_t(\theta)$ and $\bar{M}_t(\theta)$, which gives us the result. Note that $M_t(\theta)$ is motivated by the above (*) and $\bar{M}_t$ brings in λI.

Note: for the intuition, we use $\tilde{V}_t = V_t$ (when $\lambda = 0$) is invertible but this is not needed in the actual derivation.

$\bar{M}_t$ derivation:

$$M_t(\theta) = e^{\theta^T S_t - \frac{1}{2}\|\theta\|^2 \tilde{V}_t}$$

$$\theta \sim N(0, \lambda^{-1} I)$$

$$\Rightarrow \int \frac{(\sqrt{\lambda})^d e^{\theta^T S_t - \frac{1}{2}\|\theta\|^2 \tilde{V}_t}}{\sqrt{2\pi}} \exp\left(-\frac{\lambda \theta^T \theta}{2}\right) d\theta$$

$$= \int \frac{(\sqrt{\lambda})^d}{\sqrt{2\pi}} \exp\left(-\frac{\theta^T(\lambda I + \tilde{V}_t)\theta}{2} + \theta^T S_t\right) d\theta$$

$$= \int \sqrt{\frac{\lambda^d}{2\pi}} \exp\left(-\frac{1}{2}\left[\left\| \underbrace{(\lambda I + \tilde{V}_t)^{\frac{1}{2}}}_{V_t} \theta - \underbrace{(\lambda I + \tilde{V}_t)^{-\frac{1}{2}}}_{V_t} S_t \right\|^2 + \frac{1}{\gamma}\left\| \underbrace{(\lambda I + \tilde{V}_t)^{-\frac{1}{2}}}_{} S_t \right\|^2\right)\right) d\theta$$

$$+ \frac{1}{2} \| (\underbrace{\lambda^{-1} + V_t}) \|$$

$$= \sqrt{\frac{\lambda^d}{\det(V_t)}} \exp\left(\frac{1}{2} \| s_t \|^2_{V_t^{-1}} \right)$$